

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 411 — Past Questions & Materials
by Renance

Course: CHM 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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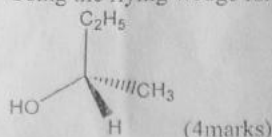
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Paper 1 of 1 — CHM 411 PQ

Level 400 • First Semester • PastQuestion

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- (b) Using the flying wedge formula, write R,S configuration of the compound below.



(4 marks)

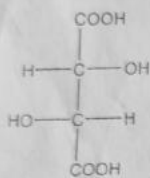
- (c) Draw and label all the Newman projection for butane. (6 marks).

- (d) Why is there no free rotation in biphenyl? (1 mark)

- 5(a) Explain the following terms:

- (i) Stereogenic center (ii) Chirality. (4 marks).

- (b) Using the Golden rule, draw and write R,S configuration for the compound below.



(5 marks).

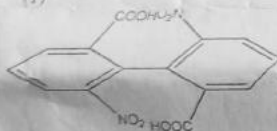
- (c) Write a short note on neighbouring group effect in S_N1 reactions. (4 marks)

- (d) Differentiate between the terms: configuration and conformation (4 marks)

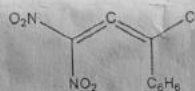
- 6(a) Write short note on optical isomerism in Spiro compounds (5 marks).

- 6b) Justify if the compounds below will undergo optical isomerism,

(i)



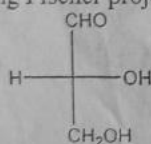
(ii)



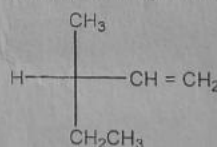
(6 marks).

- (c) Using Fischer projection give the R,S configuration the compound below.

i)



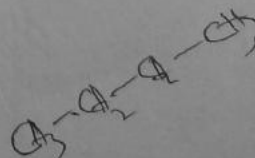
ii)



(4 marks)

- (d) Explain how the presence of OH - group on a benzene ring will affect alkylation with $\text{CH}_3\text{CH}_2\text{Br}$ / FeBr_3 (2 marks)

END OF PAPER

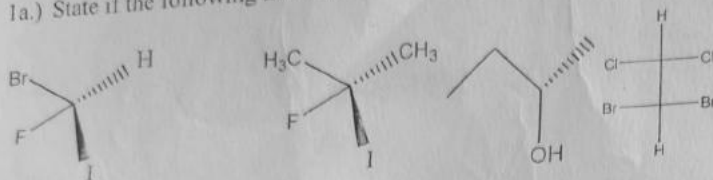


FEDERAL UNIVERSITY
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATIONS 2022/2023 SESSION

CHM 411 ORGANIC REACTION MECHANISM
Answer Any four Questions

TIME: 2½ HRS

1a.) State if the following molecules are chiral or achiral and give brief explanation (6mks)



- 1b) Draw the structure of the compound 2,4-heptadiene in the following geometrical conformer
(i). Trans, Cis (ii) Cis, Trans (2mks)
- 1c) Explain the major difference between geometrical and optical isomerism. (1mk)
- 1d) Define the following terms and illustrate with an example
(i) Enantiomers (ii) Meso compound (4mks)
- 1e) Present the following compounds in a geometrical isomeric forms
i) E-hex-2-ene (ii) cis-1,4-dipropylcyclopentane (4mks)

2a) Define stereospecific and stereoselective reactions giving an example each. (4mks)

2b) Mention three parameters that can be used to separate geometrical isomers. (3mks)

2c) Outline two importance of stereoisomeric reactions (2mks)

2d) Mention two conditions each for optical and geometrical isomerism (4mks)

2e) Define a racemic mixture and give an example (2mks)

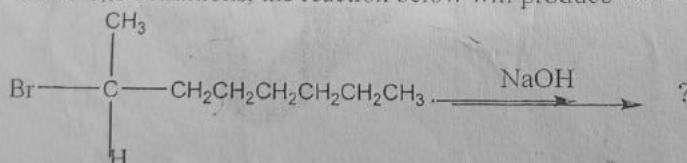
2f) Give two physical properties in which diastereomers differ. (2mks)

3a.) Give the structure of a compound with four asymmetric carbon atoms (1mk)

(i) How many stereoisomers will the compound above have? (1mk)

(ii) Draw the structures of 2-methylhexane and state if it possess chirality or not (2mks)

3b.) Under S_N2 conditions, the reaction below will produce----- (2mks)



3c.) Outline any four methods for the resolution and separation of enantiomers and explain any two. (6mks)

3d.) What are regioselective reactions? Using chemical equations only outline any two examples. (5mks)

4.(a) Write short note on the following terms as applicable in conformational analysis:

(i) Angle strain (ii) Torsional strain (iii) Van der Waals strain. (6 marks).

